KOÇ ÜNİVERSİTESİ

ELEC 205 – DIGITAL SYSTEM DESIGN
SPRING 2025

Class Meeting Location	**205 @ TBA**
Class Meeting Times	**205 @ MoWe 10:00-11:10**
PS	Th 17:30-18:40 @ SOSB07
Instructors	ENGİN ERZİN
Office Hours	TBA
Office Location	ENG 145
Office Phone	1533
Email	eerzin@ku.edu.tr
Web Address	LearnHub
Number of Credits	3
ETCS Credit	6
Prerequisites	-
Language	English
Assistant	To be announced

Course Description

Computer technology, digital hardware, Boolean algebra, logic functions and gates, canonical forms, simplification of Boolean functions, Karnaugh maps, number systems, complement arithmetic, combinational logic, adders, multiplexers, decoders, encoders, tri-state outputs, sequential logic, flip-flops, sequential circuit analysis, sequential circuit design, registers and counters, algorithmic state machines, programmable logic, central processing unit, design and simulation assignments.

Course Objectives

To teach fundamentals of digital design. To teach combinational circuit design. To teach sequential circuit design. To teach the architecture of a basic computer, including memory, datapath, control unit, and registers.

Learning Outcomes

Students will be able to

- Understand introductory digital design architectures, such as encoders, decoders, multiplexers, flip-flops, etc.
- Identify functional units and fundamental concepts of digital-computer for a given digital design task
- Design complete logic circuits

Course Contents

Starting Week	Topics
Week 1	Course regulations; Introduction to digital devices; Binary numbers; Unsigned addition/subtraction; Two's complement system;
Week 2	2's complement addition, subtraction, multiplication; BCD representation; Boolean functions; Minterm and Maxterm representations
Week 3	CLC optimization; K-Maps;
Week 4	K-Maps; PoS simplification; Don't care conditions; XOR gates;

Week 5	Half/Full adders; Decoders; Encoders; Multiplexers;
Week 6	Intro to sequential logic; Latches; D-latch; Master/Slave Flip Flops; Edge-Triggered Flip Flops;
Week 7	Sequential circuits; Edge-triggered flip-flops; SC Analysis;
Week 8	SC Design: Formulation and design steps; State assignment strategies;
Week 9	SC Design: State minimization; Other flip-flop types; Registers; Counters, Register Transfers;
Week 10	Registers, micro-operations, register transfers, shift registers; Datapath and control, non-programmable control units, algorithmic state machines (ASMs);
Week 11	ASM designs;
Week 12	Datapath and control, programmable control units;
Week 13	A simple computer architecture: Datapath, Micro-operations, Control Word;
Week 14	Review

Assessment Methods

Type	Description	Final Grade %
Attendance	Active in-class participation - 2/3 of the full attendance, will be expected for a passing letter grade.	5
Assignments	3 design and simulation assignments	15
Midterm	Assessment of student's knowledge and comprehension of the combinational circuit analysis and design, and sequential circuit analysis	35
Final Exam	Assessment of student’s knowledge and comprehension of the digital design concepts.	45
Total		100

Workload Breakdown

Type	Description	Hours
Lecture	In-class teaching	35
Tutorial	Problem-solving sessions	10
Exam	Midterm and Final exams	10
Independent Study	Final exam preparation	20
Independent Study	Midterm exam preparation	30
Independent Study	Problem-solving session preparation	10
Independent Study	Lecture preparation and review	50
Design Assignments	Design and simulation assignments	15
Total		180

Required Textbook

M.M. Mano and M.D. Ciletti, Digital Design with an Introduction to the Verilog HDL, VHDL, and SystemVerilog, 6th Edition, Pearson Education.

Make-up Policy

The midterm make-up exam will be given within two weeks after the midterm exam.

The final make-up exam will be scheduled by the Registrar and Student Affairs Directorate at the end of the semester during the period defined by the academic calendar.

The final remedial exam will NOT be offered as announced by the Registrar and Student Affairs Directorate.

Make-up exams will only be given for a valid reason (health report, etc., reaching the instructors through KUSIS or Health Center).

Academic Dishonesty

Koç University expects all its students to perform course-related activities in accordance with the rules set forth in the Student Code of Conduct[1]. Actions considered as academic dishonesty at Koç University include but are not limited to cheating, plagiarism, collusion, and impersonation. This statement's goal is to draw attention to cheating and plagiarism-related actions deemed unacceptable within the context of the Student Code of Conduct: All individual assignments must be completed by the student himself/herself, and all team assignments must be completed by the members of the team, without the aid of other individuals. If a team member does not contribute to the written documents or participate in the activities of the team, his/her name should not appear on the work submitted for evaluation.

Plagiarism is defined as "borrowing or using someone else's written statements or ideas without giving written acknowledgment to the author". Students are encouraged to conduct research beyond the course material, but they must not use any documents prepared by current or previous students, or notes prepared by instructors at Koç University or other universities without properly citing the source. Furthermore, students are expected to adhere to the Classroom Code of Conduct[2] and to refrain from all forms of unacceptable behavior during lectures. Failure to adhere to expected behavior may result in disciplinary action.

There are two kinds of plagiarism: Intentional and accidental. Intentional plagiarism (For example, Using a classmate's homework as one's own because the student does not want to spend time working on that homework) is considered intellectual theft, and there is no need to emphasize the wrongfulness of this act. Accidental plagiarism, on the other hand, may be considered as a "more acceptable" form of plagiarism by some students, which is certainly not how it is perceived by the University administration and faculty. The student is responsible for properly citing a source if he/she is making use of another person's work. For an example of accidental plagiarism, please refer to the document titled "An Example on Accidental Plagiarism". If you are unsure whether the action you will take would be a violation of Koç University's Student Code of Conduct, please consult with your instructor before taking that action.

[1] http://vpaa.ku.edu.tr/academic/studentcode-of-conduct
[2] http://vpaa.ku.edu.tr/academic/classroom-code-of-conduct